Minimum-Container Packing of a Non-Convex Polygon via Simulated Annealing and Bracketing-Bisection Search

Graduate Report, Applied Mathematics and Numerical Methods

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Abstract

A 2024 Kaggle competition posed a deceptively simple question: given N identical, irregular rigid pieces, how small a square can contain them all without overlap? Answering this exactly is NP-hard. This report describes a practical solver built around two-phase Simulated Annealing (SA) guided by an outer bracketing-bisection loop that certifiably narrows the interval containing the true minimum side length L^*. Collision detection uses the Separating Axis Theorem (SAT), a deliberate choice: SAT returns penetration depth, converting a combinatorial feasibility problem into a differentiable energy function the SA can navigate. We derive complexity estimates for the SA schedule and collision kernel, analyze how temperature governs effective step size and expected collision count, and report computational results for $N \in \{5, 10, 25, 50, 100, 200\}$ across a five-node SLURM cluster with 4 to 8 hours of wall time. Observed packing densities range from $\eta \approx 0.58$ at small N to $\eta \approx 0.48$ at $N = 200$, with feasibility becoming substantially harder to achieve as N grows.

1 Introduction

The problem is easy to state. Take a fixed non-convex polygon, make N identical copies, and find the smallest square that can hold all of them without overlap, each copy free to translate and rotate. As N grows, the configuration space expands exponentially and exact enumeration becomes hopeless.

Even the restricted case where all pieces are unit disks is NP-complete [2], as feasibility of the placement decision reduces to a variant of bin-packing. Non-convex polygons are harder still: the Minkowski sum characterizing pairwise overlap between two non-convex pieces is non-convex and potentially disconnected, so no compact feasibility certificate exists.

The method of last resort for problems of this class is stochastic local search. As Sokal observed: "Monte Carlo is an extremely bad method; it should be used only when all alternative methods are worse." That is precisely the situation here. We use Simulated Annealing, accept that it is slow and resource-intensive, and design the surrounding infrastructure to extract the most from the available compute budget.

Section 2 reviews related approaches. Section 3 states the problem mathematically. Section 4 derives complexity and runtime estimates. Section 5 develops the algorithm. Sections 6 and 7 describe the experimental setup and present results.

2 Related Work

Geometric packing appears throughout operations research, manufacturing (sheet nesting), and combinatorial geometry. The literature divides into exact methods, No-Fit Polygon heuristics, and metaheuristics.

Exact and NFP methods. The No-Fit Polygon (NFP) is the locus of positions at which one polygon just touches but does not overlap another. Bennell and Oliveira [3] survey how NFPs reduce pairwise overlap detection to a Minkowski sum computation, giving an exact characterization of the feasibility boundary. For convex polygons, the NFP is convex and computed in $O(m_1 + m_2)$ time. For non-convex polygons it is generally non-convex and possibly disconnected, requiring convex decomposition [4]. Because our polygon has genuine concavities, NFP-based exact methods are impractical at the sizes we consider.

Simulated annealing for packing. Leung et al. [5] showed that penalty-based energy functions for SA consistently outperform approaches restricted to strictly feasible states: hard feasibility constraints fragment the search space into disconnected islands, trapping local methods. Continuous penalties create navigable bridges between those islands. Jakobs [6] found that warm-starting SA from a structured initial placement substantially reduces time to feasibility.

Genetic and population-based approaches. Genetic algorithms (GAs) treat entire layouts as chromosomes and combine them via crossover. Burke et al. [7] showed that a GA encoding placement sequences discovers structurally diverse configurations simultaneously, something a single SA chain cannot do. Defining meaningful crossover operators for continuous rigid-body placements remains non-trivial, and per-generation cost scales with N in a way that makes GAs expensive beyond $N = 100$.

Parallel tempering. Parallel Tempering (PT) runs multiple SA chains at different temperatures and periodically proposes Metropolis swaps between adjacent chains [8, 9]. High-temperature chains explore freely; cold chains refine. Hukushima and Nemoto established that PT mixes exponentially faster than single-chain SA on multimodal landscapes. The present implementation uses independent multi-start SA; PT is a natural next step, though preliminary experiments under one-hour budgets found no feasible solutions under the PT variant tested, suggesting it requires substantially longer runs to benefit.

Position of this work. We adopt the penalty-energy reformulation of Leung et al., add a three-level AABB hierarchy and SAT depth computation for collision detection, and place SA inside an outer bracketing-bisection-polish loop that provides a certified bound on L^*. The combination of bisection-certified bounds with an adaptive long-run polish and deterministic HPC-safe seeding has not, to our knowledge, been applied to this specific non-convex problem.

3 Problem Formulation

Let $\mathcal{P}$ be a fixed polygon with 15 vertices and area $A_p \approx 0.2456$ (by the shoelace formula). Placing the i-th copy requires a centre $c_i = (x_i, y_i)$ and a rotation angle $\theta_i \in [0, 2\pi)$. The world vertices are $w_k^{(i)} = R(\theta_i)v_k + c_i$, where $\{v_k\}_{k=0}^{14}$ are the reference vertices and $R(\theta)$ is the standard 2×2 rotation matrix.

A configuration is *feasible* if no two copies overlap and every copy lies within $\Omega_L = [-L/2,\, L/2]^2$. The optimization problem is

$$L^* = \min L \quad \text{s.t.} \quad \exists \text{ feasible config. of } N \text{ copies in } \Omega_L. \tag{1}$$

Area lower bound. The combined area of all copies must fit inside Ω_L, giving the hard lower bound

$$L_{\text{area}} = \sqrt{NA_p}. \tag{2}$$

Every candidate $L < L_{\text{area}}$ is provably infeasible and is skipped without any SA evaluation. For $N = 100$, $L_{\text{area}} \approx 4.956$.

Why strict feasibility enforcement fails. Restricting the search to non-overlapping configurations at all times confines it to a union of disconnected feasible islands in $\mathbb{R}^{2N} \times [0, 2\pi)^N$. A local method started on any one island cannot reach another without crossing infeasible territory. Allowing temporary, penalized violations creates navigable paths between islands and is the key that makes stochastic search tractable here.

4 Complexity Analysis and Runtime Estimates

The estimates in this section use the polygon's bounding radius $r_{\text{br}} = 0.80$, area $A_p = 0.2456$, and the fan triangulation into 13 triangles. They serve both to justify design choices and to predict the fraction of wall time spent in each phase.

4.1 SA Iterations, Collision-Free

Each SA trial runs Phase A (70,000 iterations) then Phase B (35,000), totaling $M = 105{,}000$ per trial. Ignoring collision, each iteration is $O(1)$: pick a polygon, generate a displacement, apply the Metropolis criterion. The full bracket-plus-bisect phase requires at most

$$(350 + 182) \times M \;=\; 532 \times 105{,}000 \;\approx\; 5.6 \times 10^7$$

iterations (35 bracket steps at 10 trials each, 26 bisection steps at 7 trials each). At one nanosecond per iteration, this is under one minute. Collision detection, not arithmetic, is the bottleneck.

4.2 Collision Detection Cost Per SA Move

When polygon k moves, only its interactions with grid neighbours are recomputed. Let $\eta = NA_p/L^2$ be the packing density. The grid cell size is $h = 2r_{\text{br}} = 1.6$, and a search radius of $R = 2$ covers a 5×5 block of 25 cells. The expected number of neighbour polygons per query is

$$k(\eta) \;=\; 25N \cdot \frac{h^2}{L^2} \;=\; \frac{100\, r_{\text{br}}^2\, \eta}{A_p} \;\approx\; 261\eta. \tag{3}$$

At $\eta = 0.85$, about 222 candidate pairs are queried per move. Almost all are rejected by the polygon-level AABB check: two bounding boxes of radius r_{br} can only overlap if centres are within $2r_{\text{br}}$ in both x and y, covering an area of 2.56 against the 64 area units searched, a pass fraction of 4%. Of the 222 queries, roughly 9 proceed to triangle-level inspection. For those 9 pairs, the SAT kernel (six axes, $\sim$20 operations each, 169 triangle-pair combinations in the worst case but pruned by AABB) costs approximately 2,400 operations per pair. The total per-move cost is

$$C_{\text{move}}(\eta) \;\approx\; 222 \times 4 \;+\; 9 \times 2{,}400 \;\approx\; 22{,}500 \text{ flops}. \quad (4)$$

On a single core at 10^9 flops/s, each SA move takes 20 to 30 microseconds. The bracket-plus-bisect phase therefore costs roughly $5.6 \times 10^7 \times 25\,\mu\text{s} \approx 23$ minutes per worker, leaving the majority of a four-hour job for the polish phase.

4.3 Temperature, Step Size, and Effective Search Radius

Step sizes s_{xy} and s_θ are not fixed; they adapt every 1,500 iterations based on the current acceptance rate. When the acceptance rate exceeds 55%, step sizes grow by factor 1.12; below 25%, they shrink by 0.88. At high temperature T, the Metropolis criterion accepts most proposals including those creating shallow overlaps, so acceptance stays high and step sizes expand toward the ceiling $s_{xy,\max} = 2.0$. As T falls, fewer overlap-inducing moves are accepted, the acceptance rate drops, and step sizes contract toward $s_{xy,\min} = 10^{-5}$. Temperature and effective search radius thus co-decrease automatically.

The overlap depth tolerated at temperature T follows from the Metropolis acceptance probability $e^{-\lambda d^p/T}$ for a move creating penetration depth d. Setting this equal to $1/e$ gives the characteristic tolerance:

$$d_{\text{tol}}(T) \;=\; \big(T/\lambda\big)^{1/p}. \quad (5)$$

For Phase A ($p = 1$, $\lambda = 100$), at the start $d_{\text{tol}} = 0.003$ and at the end $d_{\text{tol}} = 3 \times 10^{-6}$. The polygon has unit scale, so overlaps are tolerated only up to 0.3% of its extent even at the hottest point in Phase A. The search stays near the boundary of the feasibility set, not deep inside infeasible space.

The effective diffusion coefficient in configuration space is $D(T) \;\approx\; \frac{1}{2}\alpha_{\text{target}}\, s_{xy}^*(T)^2$ with target acceptance $\alpha_{\text{target}} \approx 0.4$. Since $s_{xy}^*(T)$ grows with T, diffusion is fast at high temperature (broad exploration) and slow at low temperature (local refinement). Phase A covers the transition from global exploration to a roughly converged placement; Phase B compresses the result into strict feasibility.

4.4 Expected Collision Count at Density η

For two polygons placed uniformly at random in Ω_L, the probability of overlap is bounded by the exclusion area over total area:

$$P(\text{overlap}) \;\le\; \frac{4\pi r_{\text{br}}^2\,\eta}{N A_p}.$$

The expected number of overlapping pairs is therefore

$$\mathbb{E}[n_{\text{ov}}] \;\le\; \frac{2(N-1)\pi r_{\text{br}}^2\,\eta}{A_p}. \quad (6)$$

At $N = 100$ and $\eta = 0.85$, equation (6) gives $\mathbb{E}[n_{\text{ov}}] \lesssim 1{,}387$, an upper bound using bounding circles rather than the true polygon geometry. In practice, at temperature T with penalty λ the equilibrium total overlap energy satisfies $E \sim T$ (Gibbs equipartition), so mean total depth $\sum d_i \sim T/\lambda \to 0$ as $T \to 0$. The algorithm drives collisions to zero as it cools.

4.5 Bisection Convergence

After k bisection steps from an initial bracket of width ΔL_0, the residual uncertainty in L^* is $\Delta L_0/2^k$. With $k = 26$, this is $\Delta L_0/2^{26} \approx 1.5 \times 10^{-8}\,\Delta L_0$. For a typical initial bracket of width 0.25, absolute precision reaches 3.7×10^{-9}. In practice the limit is the quality of SA feasibility detection at each probe, not bisection arithmetic.

5 Algorithm

5.1 Collision Detection

The polygon decomposes offline into a fan of 13 triangles from vertex 0, grouped into three clusters of sizes $(5, 5, 3)$ to enable a three-level AABB hierarchy.

Level 1: bounding circle and polygon AABB. For each candidate pair from the spatial grid, a distance-squared check against $(2r_{\text{br}})^2$ runs first, then a polygon-level AABB check requiring four comparisons. Together these eliminate roughly 96% of candidate pairs before any triangle geometry is examined.

Level 2: group and triangle AABB. Surviving pairs proceed to group-level AABB checks (nine group-pair comparisons) and then per-triangle AABB checks. Both use only interval arithmetic on precomputed boxes, with no trigonometry.

Level 3: SAT with penetration depth. SAT is applied only when the AABB levels cannot rule out contact. For two triangles A and B, SAT projects both onto the six edge-normal axes (three per triangle) and tests for a separating gap. If no gap exists, the triangles overlap and the minimum overlap width across all axes is the penetration depth $d > 0$.

Table 1: Two-phase SA schedule parameters.

Parameter	Phase A	Phase B
Iterations	70,000	35,000
T_{start}	3×10^{-1}	6×10^{-2}
T_{end}	3×10^{-4}	10^{-6}
λ_0, μ_0	10^2 (fixed)	10^3 (ramped)
Overlap power p	1	2
Reinsertion prob.	2%	0.5%
Adapt window	1,500 iterations	

This depth is what makes the energy approach work. A binary overlap test would assign equal penalty to a barely-touching pair and a deeply embedded one, giving a flat energy surface with no gradient for SA to follow. The depth d is a continuous geometric signal, larger when configurations are worse, and feeding d^p into the energy creates the directional cue that guides the algorithm toward feasibility.

5.2 Energy Function

The total energy of a configuration at container size L is

$$E \;=\; \lambda \sum_{i<j} \sum_{(a,b)\in\mathcal{T}_{ij}} d(a,b)^p \;+\; \mu \sum_i \phi_i(L), \qquad (7)$$

where $d(a,b)$ is the SAT penetration depth between triangles a and b, and $\phi_i(L)$ is the squared boundary violation for copy i. The exponent p switches from 1 in Phase A to 2 in Phase B, making the penalty stiffer near feasibility. The weights λ and μ ramp upward during Phase B (factor 1.8 every 1,200 iterations, capped at 10^8) to prevent the chain from settling into jammed configurations.

5.3 Simulated Annealing Schedule

Each trial runs two phases. Phase A is the exploration phase: high starting temperature $T_0 = 0.30$, large steps, and linear overlap penalty ($p = 1$) allow pieces to relocate broadly across the square. Phase B is the enforcement phase: temperature drops from 0.06 to 10^{-6}, penalty weights ramp, and steps shrink, compressing the layout into strict feasibility.

Temperature decreases geometrically: $T_k = T_0\beta^k$ with $\beta = (T_{\text{end}}/T_{\text{start}})^{1/M}$. Geometric cooling allocates equal logarithmic time to each temperature decade, empirically balancing exploration and convergence better than linear schedules [10].

Each move selects a random polygon and either applies a small translation with optional rotation (probability $1 - p_{\text{ri}}$) or teleports the polygon to a uniform random position within Ω_L (probability p_{ri}). Rare teleportation prevents the SA from freezing in configurations where no small perturbation helps. Energy changes are computed incrementally: only pairs involving the moved polygon are recomputed via the spatial grid.

A reheating mechanism activates when the acceptance rate falls below 2%: temperature is multiplied by 1.5 and capped at $2T_0$, preventing premature freezing in Phase A.

5.4 Outer Search: Bracketing, Bisection, and Polish

Bracketing. Starting from a grid layout with 15% padding, the algorithm shrinks L by 3% or grows it by 5% per step until both a feasible upper bracket L_{high} and an infeasible lower bracket $L_{\text{low}} \geq L_{\text{area}}$ are established.

Bisection. With the bracket established, 26 steps of binary search evaluate $L_{\text{mid}} = (L_{\text{low}} + L_{\text{high}})/2$. Before each SA run at a new L, polygon centres are rescaled by $\gamma = (L_{\text{new}}/L_{\text{old}}) \times 0.98$. This warm-start step preserves structural information from the prior feasible arrangement, reducing the time to reach feasibility at the new L.

Polish. After bisection, the algorithm runs until wall time expires, attempting $L \leftarrow L^*(1-\varepsilon)$ and updating L^* on success. The shrink step ε adapts over windows of 20 attempts to target a 35% success rate, staying within $[10^{-5}, 2 \times 10^{-3}]$. Every improvement is written to disk immediately, and a global snapshot is flushed on `SIGTERM`, so SLURM preemption loses no progress.

Algorithm 1 Outer minimisation loop

1: Initialise L from grid layout; $L_{\text{low}} \leftarrow L_{\text{area}}$
2: **Bracket:** shrink or grow L by trial until feasible L_{high} and infeasible L_{low} are found
3: **Bisect:** 26 steps, warm-start positions at midpoint, update bracket
4: **Polish:** loop until wall time; adapt ε to maintain 35% improvement rate

Parallel execution. Each SLURM worker is a fully independent process. Its seed is derived deterministically from (base seed, run ID, trial index) via SplitMix64 hashing, guaranteeing reproducibility without time-based reseeding. Workers share no runtime state; the global best is found by a post-hoc pass over all output CSV files.

6 Experimental Setup

All experiments ran on the University of Arizona HPC cluster under SLURM. Two parallel launching strategies were used.

Five-node embarrassingly parallel jobs. Each main job allocated five exclusive nodes. One controller process per node spawned as many workers as available CPUs permitted after reserving a fixed number of cores for the OS (RESERVE_CPUS). With the default setting of 64 runs per node, up to 320 independent seeds ran concurrently per job. Each worker wrote its own history log and best-geometry CSV independently; no inter-worker communication occurred during the run.

Single-node array jobs. A complementary set of SLURM array jobs ran one independent seed per array task with at most two concurrent tasks per node. Array sizes were $N = 25$ (8 tasks), $N = 50$ (6 tasks), $N = 100$ (4 tasks), and $N = 200$ (2 tasks).

Smoke tests. Validation runs for $N = 5$ and $N = 10$ used a 3-minute wall time on the five-node configuration. Their purpose was to verify the pipeline end-to-end and obtain fast reference densities at small N before committing to multi-hour allocations.

Wall time and SA schedule. Jobs for $N \in \{25, 50, 100\}$ received 4 hours; the $N = 200$ job received 8 hours. The SA schedule (Table 1) was not modified across instance sizes. At $N = 200$ each polygon therefore receives on average $M/N = 525$ move proposals per trial versus 4,200 at $N = 25$. The multi-start strategy partially compensates: with many independent seeds running simultaneously, the probability that at least one finds a good arrangement increases, even if any single chain under-explores.

Bracketing used $K_b = 10$ SA trials per candidate L; bisection used $K_b = 7$; polish used $K_p = 5$.

7 Results

7.1 Quantitative Summary

Table 2 reports the best feasible side length L^* found by the combined pool of workers for each N, together with the packing efficiency $\eta = NA_p/(L^*)^2$. The area lower bound L_{area} is given for reference. A packing with $\eta = 1$ would have zero interstitial space, which is geometrically impossible for this shape; values above 0.90 are considered excellent for non-convex polygons [5].

Three observations stand out. First, the number of workers reaching feasibility drops sharply with N: from 15 of 15 smoke-test runs at $N = 5$ and $N = 10$, to 6 at $N = 25$, and to a single run at $N = 200$. This is consistent with the exponential growth in configuration space and with the under-exploration expected when M/N move proposals per polygon per trial become insufficient.

Table 2: Best feasible side length and packing efficiency by instance size. $L_{\text{area}} = \sqrt{N \cdot 0.2456}$ is the area lower bound. Smoke-test runs ($N = 5, 10$) used 3 min wall time; all others used 4 or 8 h. "Runs feasible" counts individual workers that found at least one feasible configuration.

N	L_{area}	L^*	η	Runs feas.	t_{first} (s)	Wall
5	1.108	1.472	0.567	15	5	3 min
10	1.568	2.056	0.581	15	6	3 min
25	2.478	3.315	0.559	6	8	4 h
50	3.504	4.652	0.567	4	46	4 h
100	4.956	6.785	0.533	3	53	4 h
200	7.009	10.104	0.481	1	19	8 h

Second, the best-of-best density η decreases with N, from 0.58 at small N to 0.48 at $N = 200$, suggesting that our fixed SA budget does not fully exploit the concavities of the polygon at large scale. Third, the time to final best is very long at $N = 50$ and $N = 100$ (medians of 3,657 and 3,254 seconds respectively), indicating that most improvement comes from the polish phase rather than from bisection.

7.2 Time-Series Analysis

Figure 1 shows the evolution of the best known L^* over wall time for the legacy instance sizes $N \in \{5, 10, 25, 50, 100, 200\}$, with one row per N. Each row shows individual feasible traces per worker (thin lines) together with the aggregate best-so-far envelope (thick line).

Figure 2 separates geometry from dynamics by showing the best recovered packing configuration for each N in its own panel.

Figure 3 shows the best-so-far packing density $\eta = NA_p/L^2$ over wall time. Rapid initial improvement reflects the bisection phase locking in a feasible bracket; the subsequent slow tail is the adaptive polish making incremental gains. At large N the density gain during polish is proportionally larger, confirming that the most computationally expensive part of the search is also the most productive.

Figure 4 shows the distribution of time to final best configuration per instance size. At $N = 5$ and $N = 10$, most workers converge within a few minutes. At $N = 50$ and $N = 100$, the interquartile range spans thousands of seconds, indicating that the polish phase is essential and that runs benefit from longer wall-time allocations.

7.3 Density Scaling with N

Figure 5 shows packing efficiency η as a function of N (left panel) and as a function of $1/\sqrt{N}$ (right panel), computed from best available runs at $N = 5, 10, 20, 25, 50, 100, 200$.

A roughly linear relationship with $1/\sqrt{N}$ suggests that density follows $\eta(N) \approx \eta_\infty + c/\sqrt{N}$ for some constants η_∞ and $c > 0$: as polygons become more numerous, the achievable density approaches a limiting value from above.

8 Conclusion

The solver described here combines two-phase SA with an outer bracketing-bisection loop and an adaptive polish stage. Using SAT penetration depth rather than a binary overlap test was the choice that made the energy approach viable: depth provides the continuous gradient that steers SA toward feasibility from arbitrary initial configurations. The bracketing-bisection loop certifies L^* to absolute precision 3.7×10^{-9} after 26 steps under a 0.25-unit initial bracket, and the polish phase then uses the remaining wall time productively.

The experiments reveal a clear bottleneck at large N: the fixed iteration budget M is insufficient to adequately explore each polygon's degrees of freedom when $N \geq 100$, and the fraction of workers reaching feasibility drops from near-certain at $N \leq 10$ to a single run at $N = 200$. Scaling M proportionally to N, combined with longer wall times, is the first intervention to try.

GPU acceleration. Both the incremental pair-penalty computation and the embarrassingly parallel multi-trial structure map naturally to GPU architecture. A CUDA implementation could assign one thread block per polygon pair, keeping triangle vertex arrays in shared memory and running the SAT kernel without global memory traffic. The target platform on the cluster is an NVIDIA Tesla P100 (compute capability 6.0), where memory bandwidth governs the SAT loop throughput.

Advanced SA variants. Elite-Resample Multi-Start (ER-MS) periodically culls the weakest of R parallel chains and reinitialises them from elite configurations, concentrating the search near low-energy regions without full communication overhead. Parallel Tempering (PT) runs R chains at a geometric temperature ladder and proposes Metropolis swaps between adjacent temperatures; the swap criterion preserves the joint Gibbs distribution, and high-temperature replicas can seed cold-chain refinement with configurations near deep, narrow feasibility basins. Under the budget tested here, PT found no feasible solutions, but substantially longer runs may reveal its advantages.

Genetic algorithms. A GA treating full layouts as chromosomes and combining them via crossover could maintain structural diversity across many qualitatively different configurations simultaneously, something a single SA chain cannot do. Defining a crossover operator for continuous rigid-body placements that preserves good partial arrangements (such as a tightly packed cluster of pieces) is the main technical challenge, and a controlled GA-versus-SA benchmark on this instance family would be a productive next experiment.

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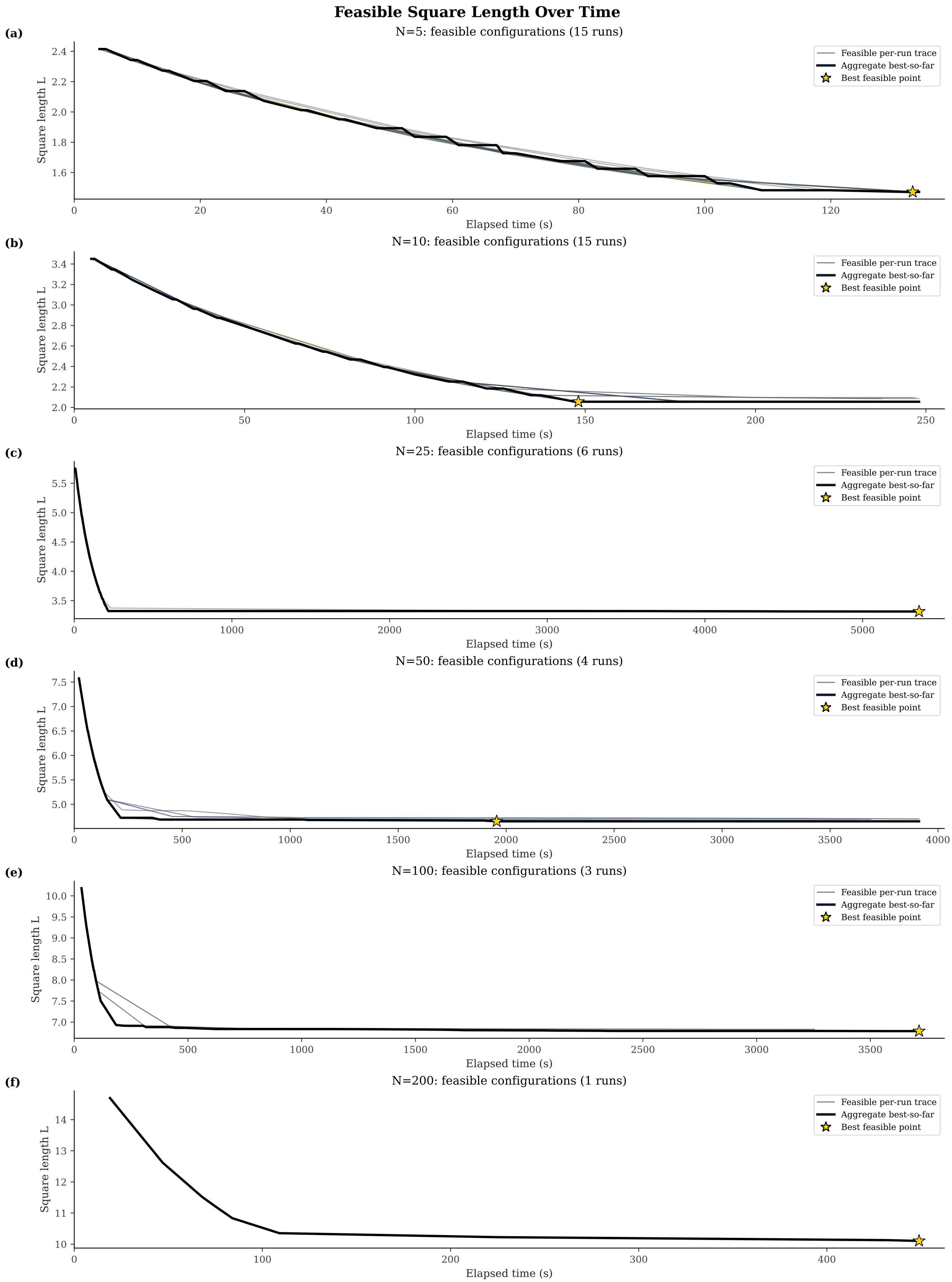

Figure 1: Feasible square length versus wall time for $N = 5, 10, 25, 50, 100, 200$, shown as rows of time-series only (no configuration inset column).

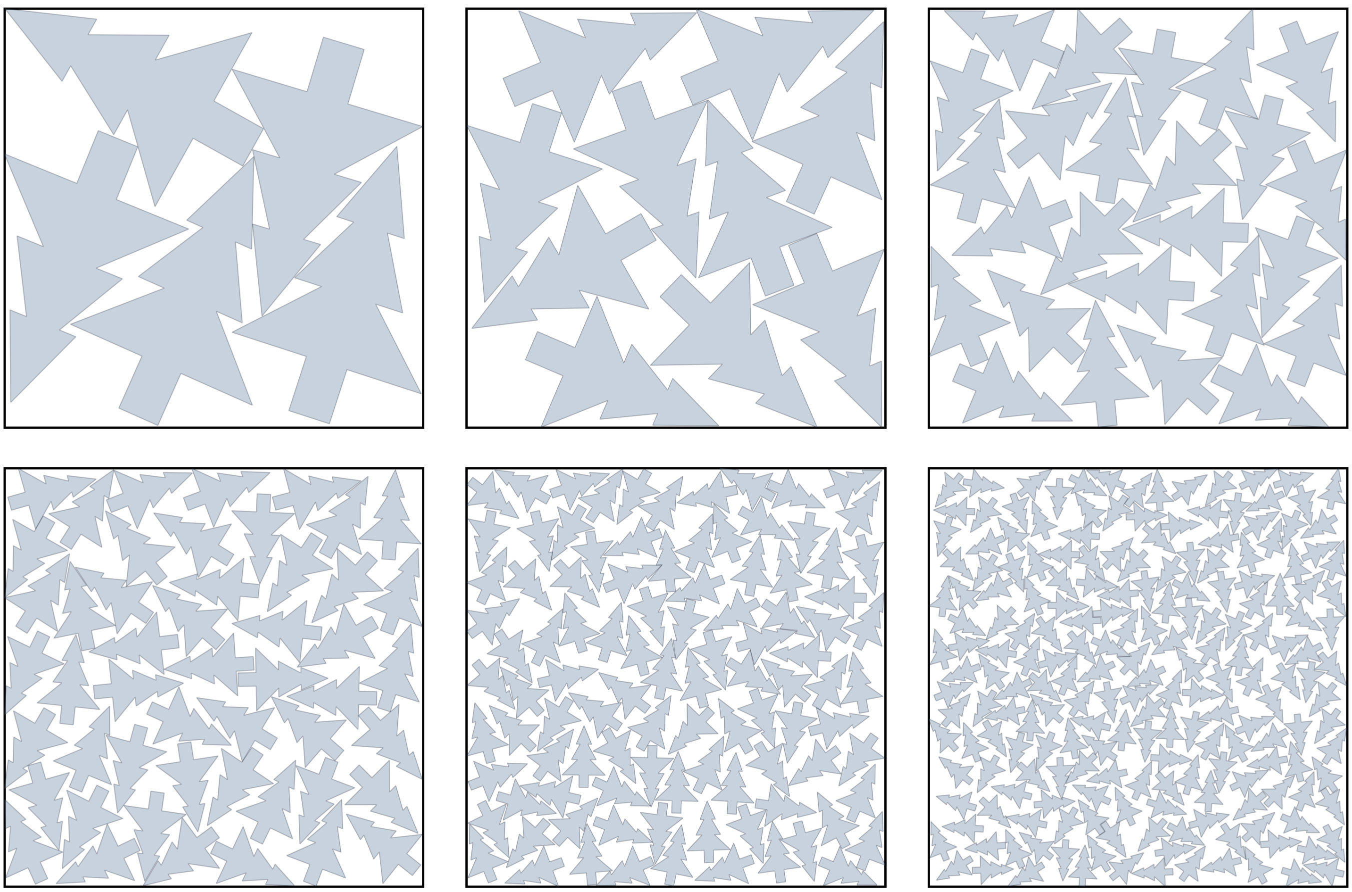

Figure 2: Best recovered packing geometries by instance size. Top row: $N = 5, 10, 25$. Bottom row: $N = 50, 100, 200$.

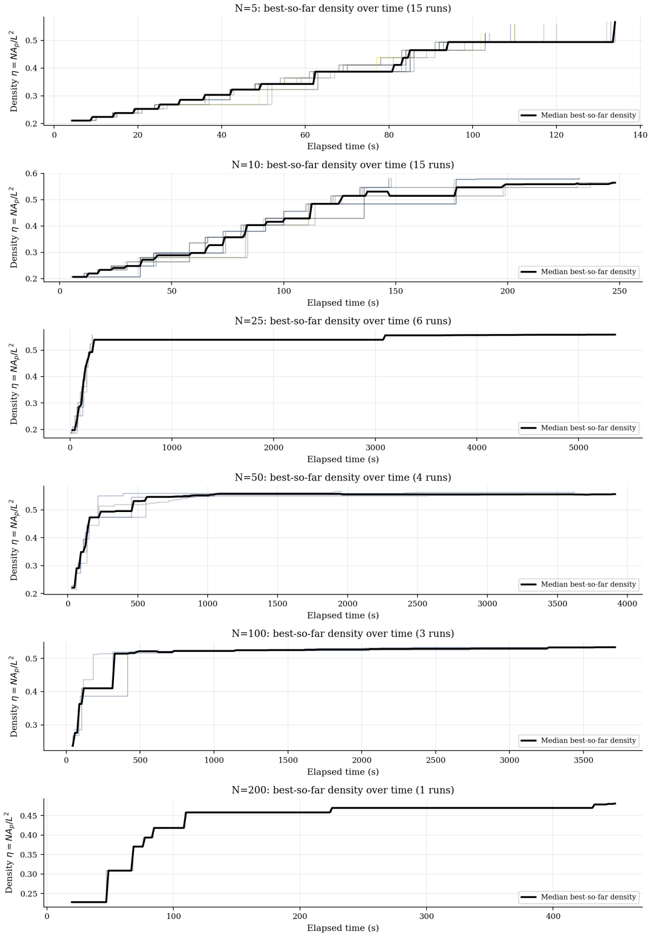

Figure 3: Best-so-far packing efficiency η versus wall time. Each panel corresponds to one instance size. Thin lines are individual worker traces; the black line is the median across workers that found feasibility.

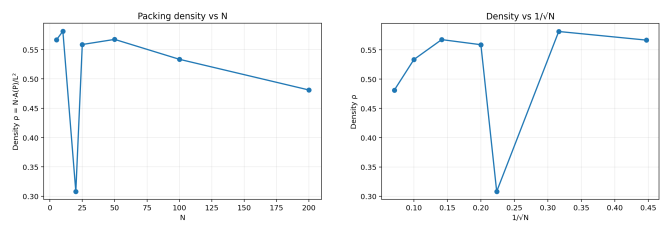

Figure 5: Packing efficiency η versus N (left) and versus $1/\sqrt{N}$ (right). The approximately linear relationship with $1/\sqrt{N}$ suggests $\eta(N) \approx \eta_\infty + c/\sqrt{N}$ with a limiting density η_∞ as $N \to \infty$.

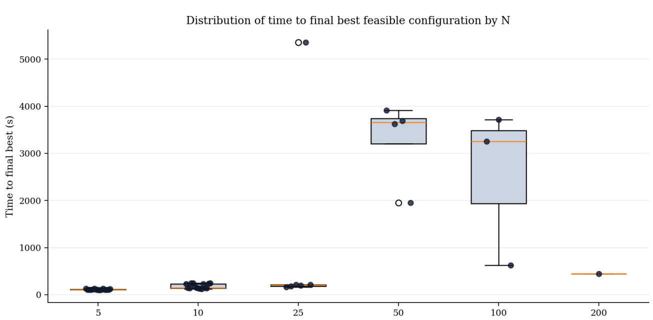

Figure 4: Distribution of wall time to final best feasible configuration by N. Boxes show interquartile range; dots are individual workers. The dramatic increase at $N \geq 50$ motivates longer allocations and more workers at large instance sizes.